

Automated E-Learning Data Pipeline

Unifying Coursera, Udemy & Udacity course data into a recommendation-ready Star Schema — Local (Airflow/dbt/DuckDB) and Cloud (Microsoft Fabric/Power BI)

Project Presentation

Agenda

- 1 Project Overview & Problem
- 2 Why It Matters for Recommendations
- 3 Architecture — Local & Cloud
- 4 Data Model (Star Schema)
- 5 Process Flow Diagrams
- 6 Challenges & Solutions
- 7 Outcomes & Results
- 8 Next Steps

The Problem: Messy, Inconsistent Course Data

Udemy

"Level 1", "300 minutes"

Coursera

"Beginner", "5 hours"

Udacity

Different tags & price formats

"Garbage In, Garbage Out" — a recommendation engine fed inconsistent data cannot accurately compare or cluster courses across platforms.

Solution: a Universal Translator & Cleaner that standardizes price, duration, level and language into one Star Schema.

Why It Matters for the Recommendation System

Cross-Platform Uniformity

Standardized level, duration & price let the model compare a Coursera and Udacity course fairly.

Feature Engineering

8 clean dimensions (instructor, domain, language, etc.) give the model rich, reliable features.

Temporal Tracking (SCD2)

Course changes are versioned, so recommendations stay tied to the syllabus a user actually saw.

Data Trust

73+ automated tests catch null IDs and broken links before they ever reach the model.

System Architecture — Two Deployment Modes

LOCAL

1. Python scrapers → CSV
2. Apache Airflow (Docker)
3. dbt-duckdb transformations
4. DuckDB warehouse file
5. Manual SQL / local BI

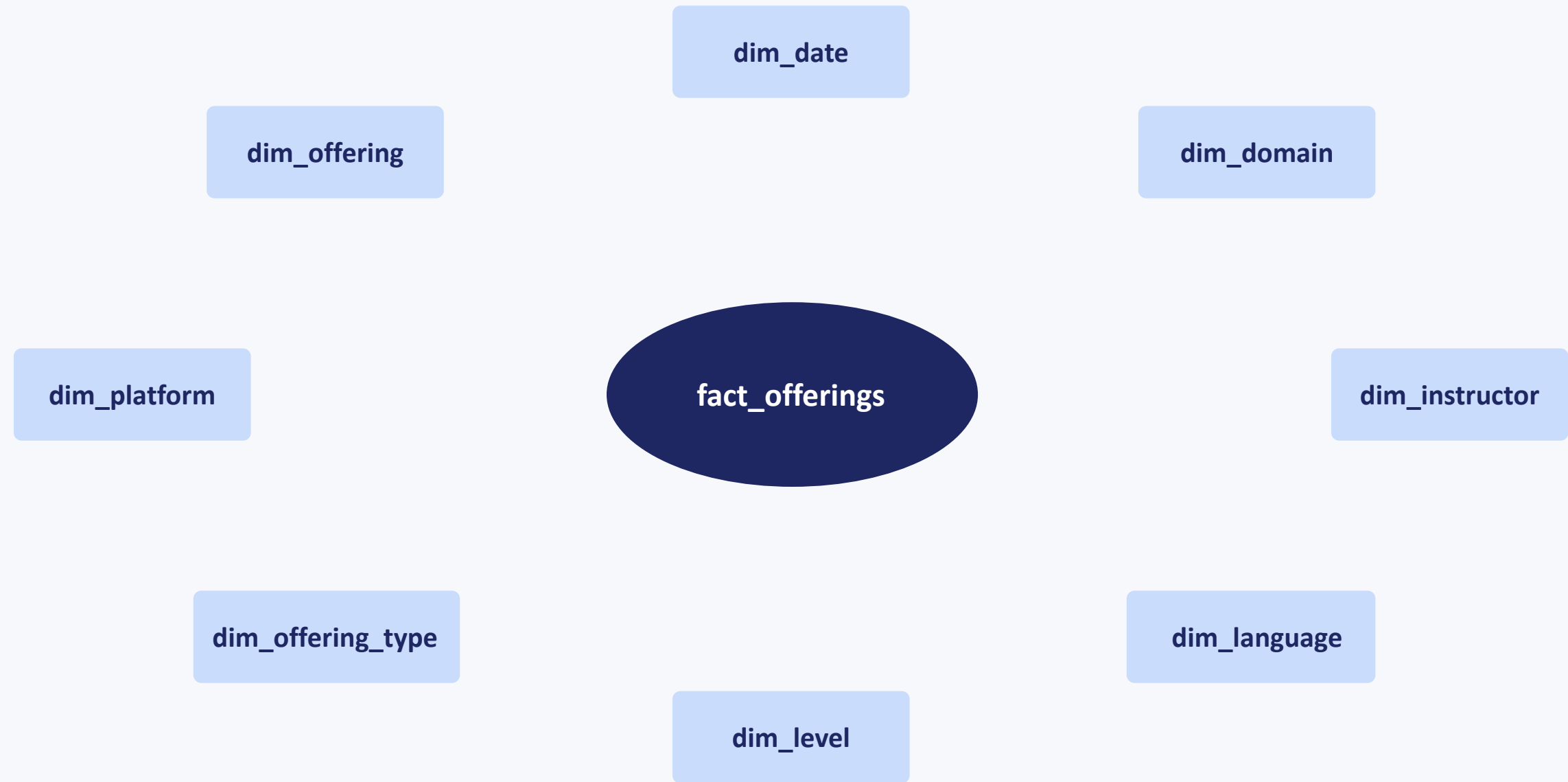
CLOUD (Microsoft Fabric)

1. Fabric Notebook ingestion
2. Fabric Pipeline orchestration
3. dbt Data Build Tool job
4. Fabric Lakehouse/Warehouse
5. Power BI semantic model + report

Technology Stack

Layer	Local	Cloud
Orchestration	Apache Airflow 2.9.3 (Docker)	Microsoft Fabric Pipeline
Ingestion	Python scrapers	Fabric Notebook
Transformation	dbt-core + dbt-duckdb	dbt (Fabric Data Build Tool job)
Database	DuckDB	Fabric Lakehouse / Warehouse
BI Layer	Local SQL / BI client	Power BI semantic model + report

Data Model — Kimball Star Schema



8 dimension tables + 1 fact table, with full SCD Type 2 history on dim_offering and 73+ dbt tests enforcing referential integrity.

Challenges & Solutions

Challenge

Inconsistent raw formats across 3 platforms

dbt build dependency cycle: snapshot needs
int_courses_standardized

Tracking historical course changes for the recommender

Guaranteeing trustworthy data for ML ingestion

Solution

Standardization macros (parse_duration, standardize_language) +
intermediate layer

Linear DAG: deps → seed → snapshot → run → test, instead of a
single 'dbt build'

SCD Type 2 snapshot (offering_snapshot) with dbt_valid_from /
dbt_valid_to

73+ automated dbt tests: not_null, unique, accepted_values,
relationships

Outcomes & Results

3

platforms unified

8+1

dims & fact tables

73+

automated data tests

2

deployment modes (Local & Cloud)

The pipeline delivers a clean, standardized, historically-tracked Star Schema that feeds the course recommendation engine in both a fast local environment and a scalable, scheduled cloud environment with Power BI reporting.

Next Steps

- Automate scheduled scraping refresh cadence
- Extend recommendation-ready semantic layer with ML feature views
- Add monitoring/alerting on dbt test failures (both environments)
- Expand platform coverage beyond Coursera, Udemy, Udacity

Thank You